

Automatic Diagnosis of Diabetic Retinopathy via Custom CNN Architecture and Transfer Learning

Maria Muñoz Pérez · Cristina [Surname] · Computer Vision — Lab 5 · April 2026

Introduction

Diabetic Retinopathy (DR) is a leading cause of preventable blindness. This work develops a binary DR/No-DR classifier from fundus photographs across two competition tracks: a *custom* CNN trained from scratch (CUSTOM) and a *fine-tuned* pretrained backbone (FINE-TUNING). Performance is measured by AUC on a held-out 1 000-image test set evaluated via Codabench.

Dataset & Preprocessing

The dataset contains 2 000 training / 500 validation / 1 000 test fundus images with a severe class imbalance (73% No-DR, 27% DR). Right-eye images are horizontally mirrored to unify anatomical orientation.

Pipeline. `CropByEye` crops the visible retinal disc via intensity thresholding. `BenGraham` local contrast normalisation [6] ($\sigma_X=10$, circular background mask) then enhances microaneurysms and vessel structure. Images are rescaled to 256 px; training augmentations add random horizontal flip, rotation ($\pm 15^\circ$), colour jitter, and a random 224 px crop. Validation uses a centre crop. All images are ImageNet-normalised ($\mu=[0.485, 0.456, 0.406]$; $\sigma=[0.229, 0.224, 0.225]$).

Custom Model (CUSTOM)

Baseline: CustomNet

A 4-block CNN with 5×5 no-padding convolutions ($3 \rightarrow 6 \rightarrow 16 \rightarrow 32 \rightarrow 64$), BatchNorm and Max-Pool per block, flattened to 6 400-d, and head $6400 \rightarrow 120 \rightarrow 84 \rightarrow \text{Dropout}(0.5) \rightarrow 1$. SGD + StepLR; AUC = 0.570.

CustomNetV2

To improve generalisation on the small training set:

- **5 blocks, 3×3 same-padding** convolutions ($3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 256$), reducing spatial size with MaxPool per block.
- **Global Average Pooling** replaces the 6 400-d flatten, giving translation invariance and a compact 256-d representation.
- **Head:** $\text{Dropout}(0.2) \rightarrow \text{Linear}(256, 128) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.4) \rightarrow \text{Linear}(128, 1)$.
- **Optimiser:** AdamW ($\text{lr}=1\text{e-}3$, $\text{wd}=5\text{e-}3$), CosineAnnealingLR ($T_{\text{max}}=50$), 50 epochs, patience=10.

Adding `WeightedRandomSampler` raises public AUC to 0.766; with `BenGraham` preprocessing and 4-pass TTA the final best is **0.786**.

Fine-tuning Model (FINE-TUNING)

ResNet18 [2] (0.698) was superseded by **EfficientNet-B0** [1] with two-stage progressive unfreezing (`features[6-8]`, then `features[5]`; AdamW $\text{lr}=3\text{e-}4$, $T_{\text{max}}=30$, label smoothing), reaching 0.770. However, severe overfitting persists: train AUC 0.919 vs. val 0.785 at best checkpoint (gap 0.134, Fig. 1).

DenseNet121 [3] — dense connections reuse features from all preceding layers within each block, providing im-

plicit regularisation on 2 000 images (standard backbone for medical imaging). Head: $\text{Dropout}(0.5) \rightarrow \text{Linear}(1024, 1)$. *Stage 1:* `denseblock4+norm5` (≈ 2.16 M trainable), AdamW $\text{lr}=3\text{e-}4$, $T_{\text{max}}=30$, patience=7. *Stage 2:* additionally unfreezes `denseblock3+transition3` at $\text{lr}=3\text{e-}5$ with a 3-epoch linear warmup (`SequentialLR`) to prevent gradient spikes; same gate (AUC ≥ 0.752) and revert logic. Result: public/private AUC **0.781/0.781** — near-zero pub-priv gap vs. 0.026 for EfficientNet.

Training Optimizations

Class-imbalance handling. `BCEWithLogitsLoss` with `pos_weight` ≈ 2.76 addresses imbalance at the gradient level. `WeightedRandomSampler` oversamples the minority class at the mini-batch level (+0.022 AUC).

Test-Time Augmentation (TTA). Scores are averaged over four passes: original, horizontal flip, vertical flip, and 180° rotation (+0.008–0.012 AUC). Vertical flip is applied at inference only (not training) to preserve the ImageNet upright prior for the fine-tuned model.

Label Smoothing. For the fine-tuned model only ($\epsilon=0.05$), hard targets become soft $\{0.025, 0.975\}$, penalising overconfident logits without damaging the limited training signal [5].

Ensemble. Grid search over $w \in \{0.10, \dots, 0.30\}$ on validation blends CustomNetV2 and fine-tuned scores for the CUSTOM submission (skipped if custom val AUC ≤ 0.62). A consistent val-test gap of ≈ 0.02 – 0.03 means gains below ≈ 0.01 on validation should not be over-interpreted.

Ablations & Negative Results

- **CLAHE:** broke `CropByEye`'s intensity threshold (-4.6% custom AUC).
- **Cache pipeline:** fixed 224 px crops removed per-epoch `RandomCrop` spatial jitter (-0.028 AUC despite $5 \times$ speed-up).
- **Scale jitter (`TVRandomResizedCrop`):** no gain (-0.004).
- **BenGraham (buggy):** assumed float $[0, 1]$; `CropByEye` passes uint8 $[0, 255]$ — `clip(uint8, 0, 1)` collapsed all pixels to $\{0, 1\}$, producing uniform grey (custom 0.541, ft 0.558). Fixed with dtype guard and circular mask: custom private +0.027 (0.743 $\rightarrow$ 0.770).

Individual Contributions

Cristina [Surname]: Baseline exploration in `first_attempt.ipynb` — class-imbalance analysis, CustomNet design, initial data pipeline, and first Codabench submission.

Maria Muñoz: All subsequent work (experiments B–I) — CustomNetV2 design, EfficientNet-B0 two-stage fine-tuning, `WeightedRandomSampler`, 4-pass TTA, ensemble grid search, label smoothing, `BenGraham` preprocessing (dtype fix + circular mask), DenseNet121 two-stage fine-tuning with warmup scheduler (I-1/I-2), and all notebook documentation.

Supplementary Material: Tables, Figures and References

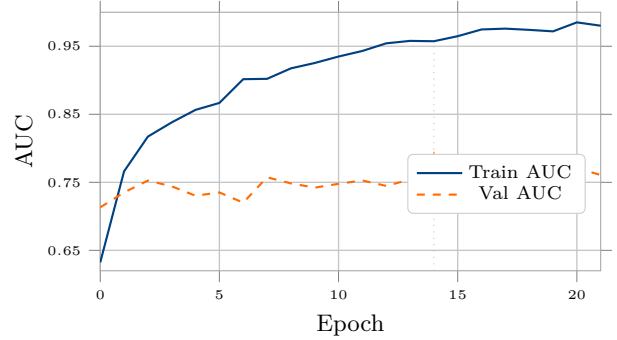
Table 1: AUC results for all submitted experiments (*pub* = public leaderboard, *priv* = private). **Bold:** best result per category. *Italic / red rows:* discarded or failed experiment.

Notebook	Key change(s)	Custom AUC		FT AUC	
		pub	priv	pub	priv
first_attempt	CustomNet (baseline) + ResNet18 fine-tuning	0.570	0.597	0.698	0.708
attempt_with_b	CustomNet: Adam optimiser + CosineAnnealingLR	0.558	0.566	0.692	0.693
attempt_with_a_b	<i>CLAHE preprocessing — broke CropByEye (−4.6%)</i>	<i>0.544</i>	<i>0.562</i>	<i>0.711</i>	<i>0.692</i>
attempt_with_c	FT: switch to EfficientNet-B0 + AdamW	0.561	0.579	0.751	0.734
second_attempt	CustomNetV2; 2-stage progressive FT; TTA; Ensemble	0.744	0.731	0.742	0.727
second_attempt_tuned	+ WeightedRandomSampler (balanced mini-batches)	0.766	0.743	0.770	0.744
s.a.t._cache	<i>Cache pipeline — removes RandomCrop jitter (−0.028)</i>	<i>0.736</i>	<i>0.730</i>	<i>0.742</i>	<i>0.728</i>
third_attempt	Scale jitter (ResizedCrop); label smoothing; $T_{\max}=25$	0.762	0.742	0.766	0.743
fourth_attempt	<i>BenGraham — uint8 dtype bug, all images grey (−22%)</i>	<i>0.541</i>	<i>0.557</i>	<i>0.558</i>	<i>0.533</i>
fifth_attempt	BenGraham (dtype-fixed) + 4-pass TTA + label smoothing	0.770	0.770	0.757	0.755
rowbest sixth_attempt	DenseNet121 (I-1/I-2) + warmup scheduler — new best	0.786	0.787	0.781	0.781

Table 2: Architecture and training comparison.

Feature	CustomNet	CustomNetV2
Conv blocks	4 (5×5, no-pad)	5 (3×3, same-pad)
Channels	3→6→16→32→64	3→32→64→128→256→256
Spatial head	Flatten (6400-d)	Glob. Avg Pool (256-d)
FC head	6400→120→84→1	256→128→1
Conv Bias	Yes	No (BN absorbs)
Optimiser	SGD	AdamW (wd=5e-3)
LR schedule	StepLR ($\gamma=0.1$)	CosineAnnealingLR
Sampler	Uniform	Weighted (class)
Preprocessing	None	BenGraham
TTA	None	4-pass
Public AUC	0.570	0.786

Figure 1: EfficientNet-B0 Stage1 training curves (second_attempt_tuned). Train AUC reaches 0.957 at the best val checkpoint (*, epoch 14) while val AUC peaks at 0.779, revealing a gap of 0.178. This persistent overfitting motivated the switch to DenseNet121 in **sixth_attempt**, which eliminated the gap (public and private AUC both 0.781).



References

- [1] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proc. ICML 2019*.
- [2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proc. CVPR 2016*, 770–778.
- [3] Huang, G., Liu, Z., van der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. *Proc. CVPR 2017*, 4700–4708.
- [4] Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *Proc. ICLR 2019*.
- [5] Müller, R., Kornblith, S., & Hinton, G. (2019). When does label smoothing help? *Proc. NeurIPS 2019*, 4694–4703.
- [6] Graham, B. (2015). Kaggle diabetic retinopathy detection competition report. *University of Warwick*.